

Mapping neuro-vascular unit communications reveals distinct angiogenic programs across developing mouse brain regions

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Neurovascular unit (NVU) communications guide vascular patterning, BBB maturation, and neuronal homeostasis, yet whether these interactions differ across brain regions and how they regulate the vasculature during development remains unclear. Here, we combine spatial transcriptomics with region-resolved endothelial single-cell RNA sequencing in the postnatal mouse brain to map cortical and thalamic NVU communication dynamics. We uncover spatiotemporal divergence of endothelial programs and show that neuronal and glial maturation parallels region-specific angiogenic trajectories. We identify neuronal–endothelial TGF β 2 signaling as an essential regulator of thalamic vascularization during a defined postnatal developmental window. Loss of endothelial TGF β R1 signaling leads to mTOR hyperactivation and thalamus-predominant vascular malformations and hemorrhage within this developmental window, and these defects are rescued by mTOR inhibition. Together, these findings show that circuit maturation and region-specific NVU communication programs coordinate postnatal angiogenesis and vascular maturation, providing a framework for understanding regional vulnerability in neurovascular disorders.

Local signals and cellular interactions between blood vessels and the surrounding microenvironment are crucial for promoting vascular specification and shaping organotypic vascular networks, both essential for optimal tissue function^{1–4}. In the brain, blood vessels and the surrounding neurons, glial cells, and perivascular cells (pericytes

and smooth muscle cells) together form the neurovascular unit (NVU). Communication within the NVU governs vascular patterning, blood–brain barrier (BBB) formation and maintenance, and neurovascular coupling, thereby ensuring neuronal homeostasis and higher brain function^{5,6}.

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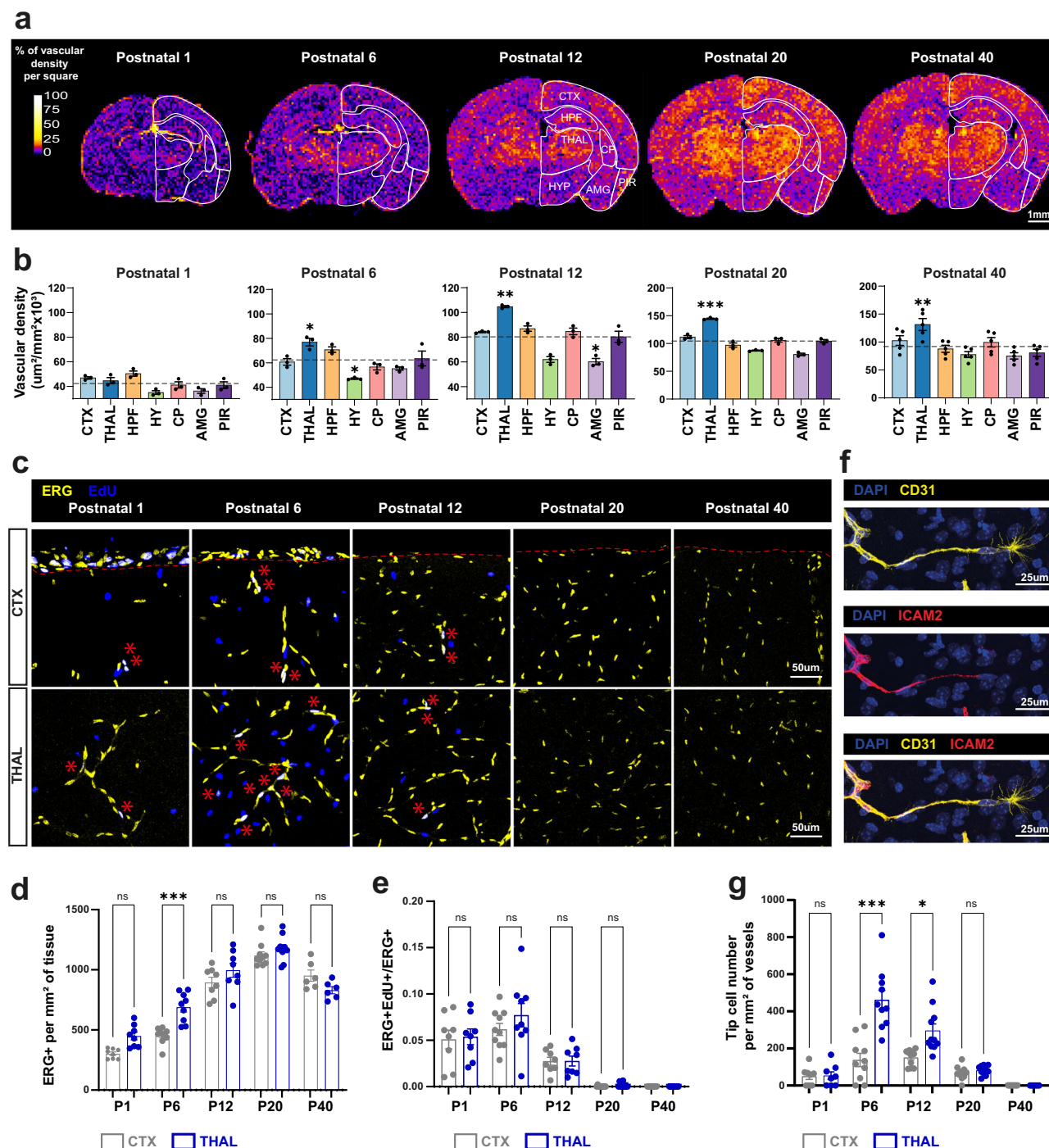


Fig. 1 | Regional vascularization dynamics in the postnatal brain. a Heatmap representation of the brain vascular density of 7 brain regions (CTX: cortex; HPF: hippocampal formation; THAL: thalamus; HY: hypothalamus; PIR: piriforme; CP: caudoputamen; AMG: amygdala) based on CD31 and ICAM2 immunostaining (the heatmap is representative of $n=3$ mice for P1, P6, P12, and P21 and $n=5$ mice for P40). **b** Quantification of the vascular density ($\mu\text{m}^2/\text{mm}^2 \times 10^3$). The dotted line represents the total brain vascular density average at each time. The dot plots are representative of $n=3$ mice for P1, P6, P12, and P21 and $n=5$ mice for P40. (mean P6 vs. THAL P6 $p=0.0306$, mean P6 vs. HY P6 $p=0.0437$, mean P12 vs. THAL P12 $p=0.0066$, mean P12 vs. AMG P12 $p=0.0279$, mean P20 vs. THAL P20 $p=0.0006$, mean P40 vs. THAL P40 $p=0.0047$). **c** Double ERG and EdU immunolabeling. The red dotted line delineates the boundary between the pia mater and the brain parenchyma. The red asterisks indicate the EdU proliferative ECs. **d, e** Quantification of brain parenchyma ERG⁺ cells per mm^2 of tissue (P6 CTX vs. P6 THAL $p=0.0002$)

and ERG⁺EdU⁺/ERG⁺ cells (**e**). **f** Representative picture of DAPI, CD31, and ICAM2 immunostaining of tip cell. Note that the tip cell has CD31/ICAM2 filopodia. **g** Quantification of tip cell number per mm^2 of vessels (dot plots represent pictures from at least 3 brains per condition and area, P6 CTX vs. P6 THAL $p=0.0004$, P12 CTX vs. P12 THAL $p=0.0121$). Values represent means $\pm$ s.e.m, P values ($*P < 0.05$, $**P < 0.01$, $***P < 0.001$) were calculated by ordinary one-way ANOVA with Dunnett's multiple comparisons test (**b**) and ordinary one-way ANOVA with Tukey's multiple comparisons test (**d**), Kruskal-Wallis with Dunn's multiple comparisons test (**e**) or Brown-Forsythe and Welch ANOVA with Dunnett's T3 multiple comparisons test (**g**), and P values were calculated by one-way ANOVA using a comparison to the average vascular density in (**b**). All statistical tests were two-sided. Source data are provided as a Source Data file.

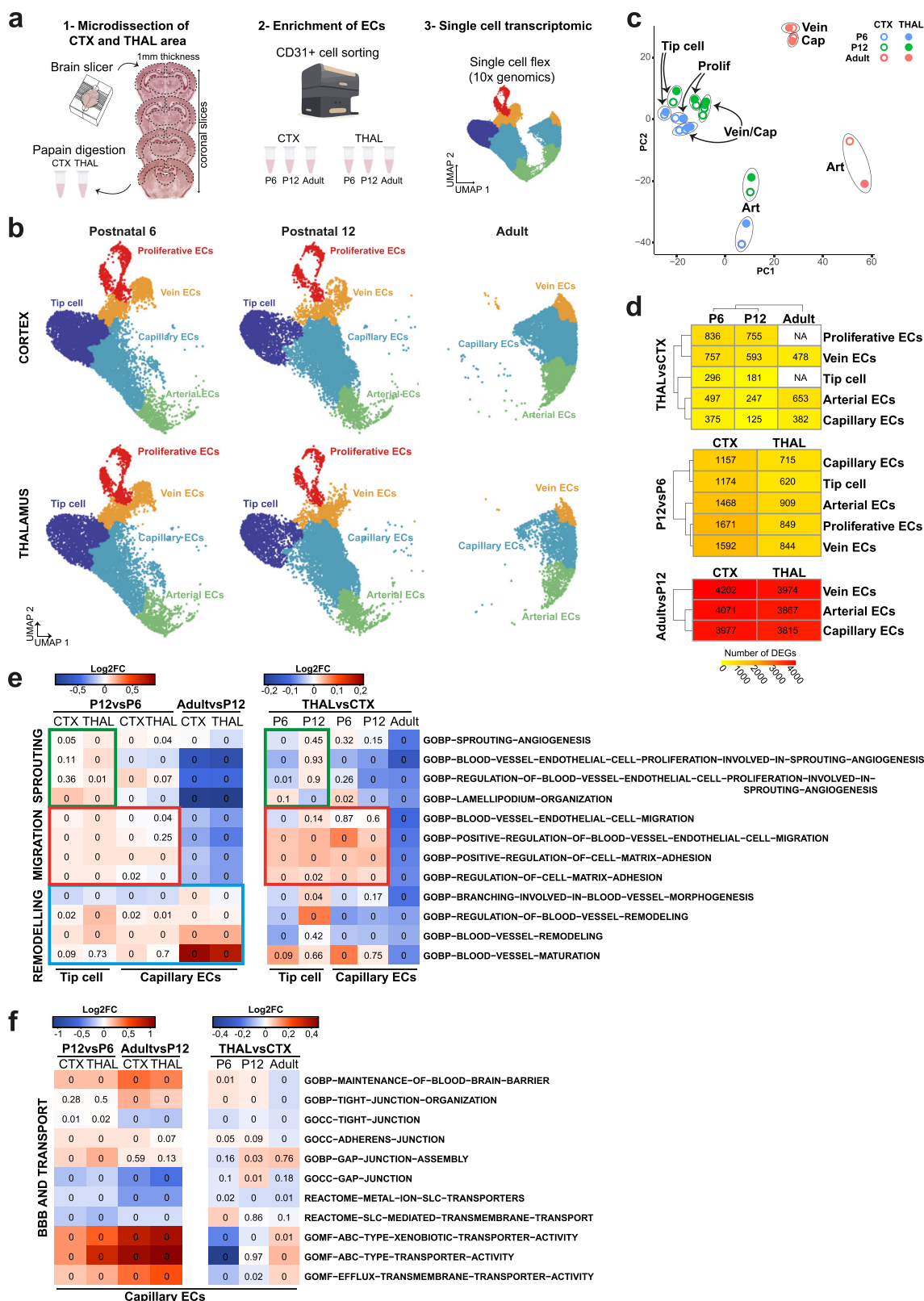


Fig. 2 | Endothelial heterogeneity in cortex and thalamus across postnatal development. **a** Schematic of the experimental strategy for region-resolved endothelial scRNAseq from CTX and THAL at P6, P12, and adult stages. **b** UMAP representation of all ECs from CTX and THAL across stages. **c** PCA plot of CTX and THAL ECs based on averaged single-cell expression. **d** Heatmap of differentially expressed genes across EC subtypes, regions, and age. **e, f** Heatmaps of

differentially expressed angiogenic (**e**), BBB-related, and transporter (**f**) pathway activities across EC subtypes between regions and developmental stages, with respective *p*-values written on heatmaps calculated by the Wilcoxon Rank Sum test. (UMAP Uniform Manifold Approximation and Projection, PCA Principal Component Analysis, EC endothelial cell, CTX cortex, THAL thalamus). Source data are provided as a Source Data file.

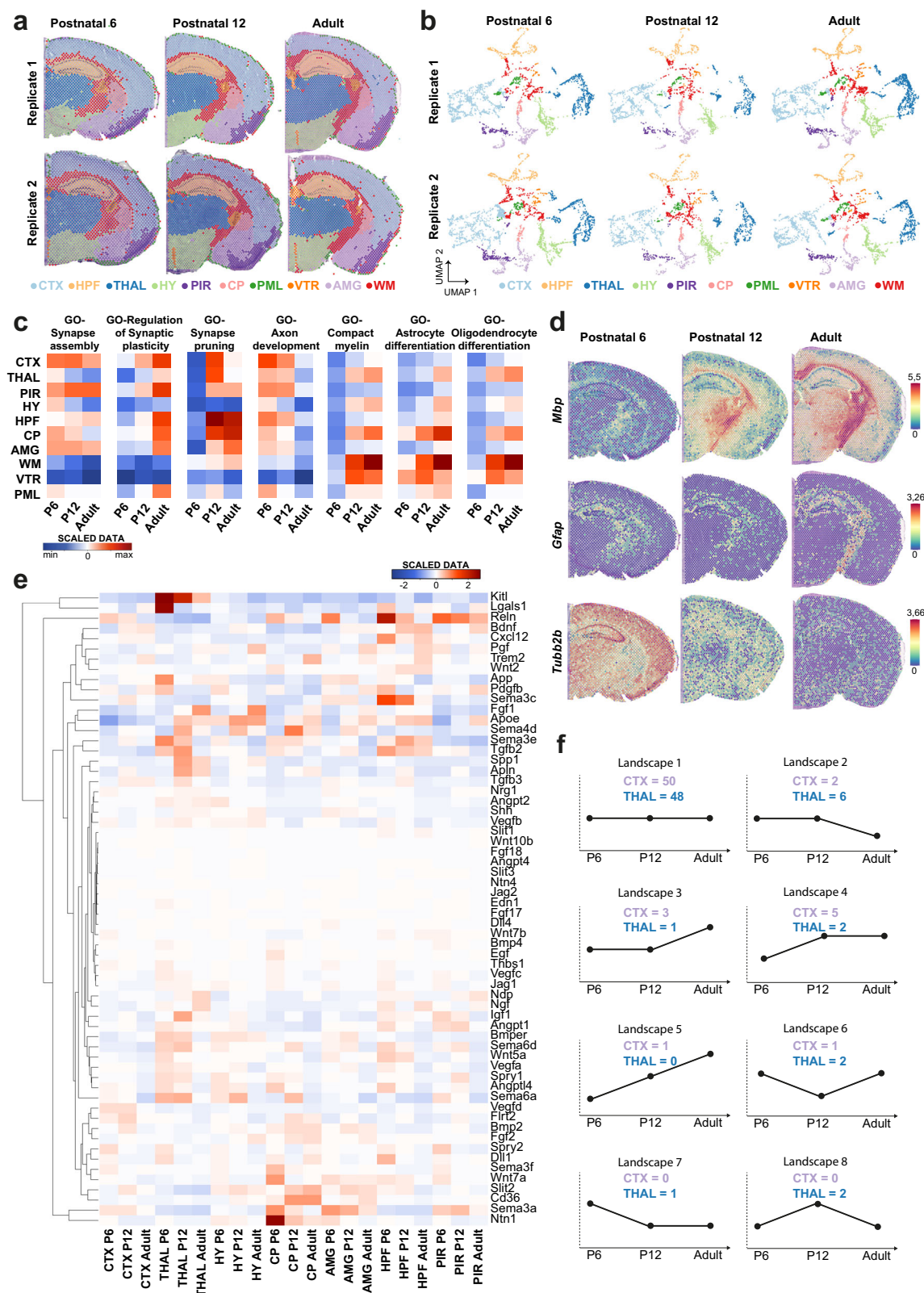


Fig. 3 | Spatial profiling of key angiogenic signatures across developing brain regions. a Visium plot of P6, P12, and adult brain regions of replicate 1 and 2 (CTX; HPF, THAL; HY; PIR; CP; AMG; PML: pia mater like; VTR: ventricles; WM: white matter). **b** UMAP plot of P6, P12, and adult brain regions of replicate 1 and 2. **c** Heatmap representation of selected pathway expression based on scaled pathway activity score (calculated with VISION package) between P6, P12, and adult brain regions. **d** Visium plot of *Mbp*, *Gfap* and *Tubb2b* expression in P6, P12 and adult

brain regions. **e** Heatmap representation of the scaled expression of angiogenic cues between P6, P12 and adult brain regions and **f** their representative expression profiles across time in the CTX and THAL brain regions. (CTX cortex, HPF hippocampal formation, THAL thalamus, HY hypothalamus, PIR piriforme, CP caudoputamen, AMG amygdala, PML pia mater like, VTR ventricles, WM white matter). Source data are provided as a Source Data file.

Recent studies have shown that the adult brain exhibits significant vascular differences, including heterogeneity in BBB permeability, vessel density, diameter, and branching patterns, alongside diversity in neuronal and glial cell populations across regions^{7–14}. These observations suggest that distinct spatiotemporal cues arising from NVU cell–cell communications may promote regional differences in angiogenesis and BBB development and maintenance. To date, most studies of brain angiogenesis have focused on forebrain regions during embryogenesis. The development of the forebrain's vascular network starts at the embryonic day 10.0–11.5 (E10.0–E11.5) and expands throughout postnatal development until postnatal day 25 (P25). Simultaneously, ECs develop a BBB to control exchanges between the bloodstream and the neuronal tissue. Recent studies have revealed an unnoticed second wave of angiogenesis in the developing cortex during postnatal development^{15–17}. EC proliferation rapidly increases from P5 and peaks around P10 to drop drastically at P15, whereas vessel branchpoints and length increase until P25, underscoring the role of sprouting tip cells during this secondary angiogenic wave¹⁷. This second postnatal angiogenic wave is regulated by neuronal activity and oligodendrocyte maturation^{15,16,18–20}, suggesting region-specific and developmentally dynamic NVU communications. However, it remains unclear whether and how region-specific signals and NVU cell interactions regulate vascular patterning during postnatal brain angiogenesis.

In this study, we aimed to investigate the molecular mechanisms regulating regional vascular heterogeneity during postnatal brain development. To achieve this, we combined spatial transcriptomics with region-resolved endothelial single-cell RNA sequencing in the postnatal mouse brain, revealing that endothelial cells already display distinct angiogenic and barrier-associated transcriptional programs across brain regions during development. We then used these datasets to map NVU communications and identify cortical and thalamic-specific ligand–receptor interactions. Among them, transforming growth factor β (TGF β) emerged as a regulator of vascular sprouting and integrity in the thalamic area during an early postnatal developmental window. Altogether, our NVU communication maps refine our understanding of brain vascularization and provide a framework to interpret regional vulnerability in neurovascular diseases and guide future therapeutic strategies.

Results

Regional vascularization dynamics in the postnatal brain

To assess postnatal regional heterogeneity in the cerebral vasculature, we first quantified vascular density in seven major brain regions, such as cortex (CTX), thalamus (THAL), hippocampal formation (HPF), hypothalamus (HY), caudo-putamen (CP), amygdala (AMG), and piriform (PIR), using CD31 and ICAM2 immunostaining on coronal sections from P1 to P40 C57BL/6J mice (Fig. 1a, b and Supplementary Fig. 1a–g). As reported for the CTX^{15–17}, vascular density increased from birth to P20 across all regions (Supplementary Fig. 1h). Comparing each region to the whole-brain average revealed significant heterogeneity beginning at P6, with the THAL consistently showing the highest vascular density from P6 to P40 (Fig. 1a, b).

To explore the cellular basis of these regional differences, we quantified proliferating endothelial cells (ECs) and sprouting tip cells in CTX and THAL (Fig. 1c–g and Supplementary Fig. 2a–e). ERG⁺ ECs per mm² of tissue increased from P1 to P20, and EdU⁺/ERG⁺ proliferating ECs were detected until P12, peaking at P6, in both CTX and THAL (Fig. 1c–e and Supplementary Fig. 2c, d). However, ERG⁺ ECs per mm² of vessel showed a decrease of endothelial density in THAL compared to CTX between P12 and P40, suggesting a more pronounced vascular remodeling (Supplementary Fig. 2c). CD31⁺/ICAM2^{low}/filopodia⁺ tip cells were also present from P1 to P20, with peak abundance at P6 (Fig. 1f, g, and Supplementary Fig. 2a, b, e). Notably, tip cells were more abundant in THAL than CTX at both time

points, suggesting stronger tip cell–driven angiogenic activity in the thalamus and likely contributing to its higher vascular density.

Together, these results reveal both temporal and regional differences in vascular density within the postnatal mouse brain. After P12, EC proliferation decreases sharply, whereas tip cell activity persists, suggesting a transition from a proliferative phase to tip cell–driven vascular remodeling.

Endothelial heterogeneity in cortex and thalamus across development

Given the distinct vascular features of the THAL and CTX, we profiled ECs from both regions using single-cell RNA sequencing at P6, P12, and adult stages (Fig. 2a). Sub-clustering and annotation based on established EC markers^{13,21,22} identified venous (*Myo1^{high}*, *Adh1^{high}* and *Adgrg6^{high}*), proliferative (*Hmml^{high}*, *Klf2c^{high}* and *Eso2^{high}*), capillary (*Cdkn2b^{high}* and *Serpine2^{high}*), arterial (*Bmx^{high}*, *Fbln5^{high}* and *Tmem100^{high}*), and tip cell (*Chst1^{high}* and *Apod^{high}*) EC clusters (Fig. 2b and Supplementary Fig. 3a). Cell numbers for each cluster and stage are reported in Source Data file.

Uniform Manifold Approximation and Projection (UMAP) analysis showed that P6 and P12 ECs cluster closely, whereas adult ECs shift markedly, reflecting a major transcriptional transition as vessels mature (Fig. 2b). Proliferative and tip cell clusters were present at both P6 and P12 but not in the adult stage, with a sharper reduction in proliferative ECs than tip cells, consistent with Fig. 1 (Supplementary Fig. 3b). Tip cells expressed the D-tip marker *Apod* but not the S-tip marker *Esm1*, indicating a neuronal D-tip identity as in the neuroretina²² (Supplementary Fig. 3c). PCA and differential gene expression across regions, stages, and EC subtypes (Fig. 2c, d and Supplementary Data 1) revealed that temporal differences dominated over regional ones, though mildest regional specificity remained detectable.

Pathway enrichment analysis reveals that while sprouting of tip cells shows a slight increase at P12 compared to P6, it appears similar between CTX and THAL (Fig. 2e). Interestingly, thalamic tip and capillary ECs exhibit higher migratory potential than their cortical counterparts, which may contribute to the vascular differences observed between these regions. Vascular remodeling pathways progressively increased in both regions toward adulthood, reflecting maturation of vascular architecture (Fig. 2e).

Our analysis also revealed dynamic BBB maturation during postnatal development. While junctional and barrier-establishing pathways progressively increased from P6 to adulthood, transport-related programs showed more complex trajectories. While several transporter activities declined with age (REACTOME-SLC-MEDIATED-TRANSMEMBRANE-TRANSPORT and REACTOME-METAL-ION-SLC-TRANSPORTERS), others increased in adulthood (GOMF-ABC-TYPE-XENOBIOTIC-TRANSPORTERS-ACTIVITY, GOMF-ABC-TYPE-TRANSPORTER-ACTIVITY and GOMF-EFFLUX-TRANSMEMBRANE-TRANSPORTER-ACTIVITY), reflecting a progressive diversification and specialization of BBB transport functions (Fig. 2f). Although most permeability- and transport-related changes followed similar developmental dynamics in THAL and CTX, we also observed a few region-specific differences, suggesting that BBB functional maturation is not entirely uniform across the brain.

Regional transcriptomic programs underlying developing brain angiogenesis

To gain further insight into the molecular cues underlying these spatiotemporal differences, including the distinct angiogenic dynamics and the higher migratory transcriptomic signature of thalamic ECs, we profiled regional gene programs using spatial transcriptomics (Visium, 10X Genomics) at P6, P12, and adult (2 months) in C57BL/6J males ($n = 2$ per age) (Fig. 3a and Supplementary Fig. 4a, b). P6 and P12 represent two distinct angiogenic stages characterized by endothelial

proliferation and sprouting, whereas adults provide a quiescent reference. Using spatial and UMAP clustering, established regional markers and the Allen Brain Atlas, we consistently annotated ten regions across ages, such as CTX, HPF, THAL, HY, CP, AMG, PIR, white matter (WM), ventricles (VTR), and pia mater-like (PML) domains (Fig. 3a, b and Supplementary Fig. 4c). UMAP representation showed that regional spots cluster together independently of age, confirming our annotations and robust regional transcriptomic similarities underlying brain anatomy.

We analyzed the differentially expressed genes (DEG) between brain regions across timepoints (Supplementary Fig. 4c and Supplementary Data 2), and we could identify well-known markers specifically enriched in brain regions, such as *Mef2c*, *Tcf7l2*, and *Zbtb20* expression in the CTX, THAL, and HPF, respectively (Supplementary Fig. 4d, e). We confirmed our findings using the in-situ hybridization (ISH) database of the developing mouse brain from the Allen Institute (Supplementary Fig. 5a–c, <https://developingmouse.brain-map.org/>).

Pathway analysis recapitulated well-known postnatal transitions of the critical periods characterized by neuronal activity-induced synaptic remodeling and glial cell differentiation over time (Fig. 3c). Axon development and synapse assembly signatures decreased after P12, whereas regulation of synaptic plasticity and synaptic pruning increased during brain development. Moreover, myelin, astrocyte, and oligodendrocyte differentiation signatures increased from P12 onward (Fig. 3c). We confirmed an increase in mature oligodendrocyte and astrocyte marker expression from P12 using Visium plots, such as *Mbp* and *Gfap*, respectively (Fig. 3d and Supplementary Fig. 4f). We also detected a decrease in *Tubb2b* and an increase in *Rora*, reflecting axon development and neuronal maturation (Fig. 3d and Supplementary Fig. 4d–f). As expected, WM exhibited low synaptic and high oligodendrocyte signatures. However, because each Visium spot captures a group of cells, we cannot determine whether these transcriptomic shifts reflect changes in cell identity or population.

Strikingly, the developmental changes in neuronal and glial maturation signatures between P6 and P12 parallel the vascular dynamics described in Fig. 1, suggesting distinct region-dependent relationships between brain cell maturation and vascular remodeling.

We next profiled angiogenic ligands using an in-house curated gene set^{1,23} (Fig. 3e). Although most growth factors remained stable from P6 to adulthood, subsets of ligands displayed pronounced regional and temporal shifts. In CTX and THAL, we delineated eight distinct angiogenic expression landscapes (Fig. 3f), revealing that angiogenic factor expression undergoes region-specific and stage-dependent shifts during development. These findings indicate that the angiogenic niche is highly dynamic and shaped by distinct molecular programs across brain regions.

Identification of neurovascular unit communications in the developing brain

To determine whether region- and age-specific angiogenic ligands engage their corresponding endothelial receptors, we performed unbiased cell–cell communication analysis using NicheNet²⁴ at each time point, comparing spatial ligand expression from CTX and THAL (senders) with their respective endothelial scRNA-seq receptor profiles (receivers). We identified common and region-enriched ligand–receptor interactions across P6, P12, and adult stages (Fig. 4a and Supplementary Fig. 6a; Supplementary Data 3). VEGFA ranked across all stages, consistent with its established roles in angiogenesis, vascular homeostasis, and neurogenesis²⁵ (Supplementary Fig. 6a, b). Notably, predicted ligand–receptor interactions change from P6 to adulthood, aligning with the neuronal, astrocytic, and oligodendrocyte maturation signatures revealed by spatial transcriptomics. These parallel developmental trajectories suggest that neural–glial maturation actively shapes regional endothelial remodeling.

Region-specific interactomes revealed *Fgf9* in THAL and *Reln* in CTX, and notably, TGF β signaling was strongly enriched in the thalamus at P6–P12 (Fig. 4a, b), consistent with its known functions in tip-cell migration, angiogenesis, and vascular maturation^{22,26}. Ligand-activity scores confirmed active TGF β signaling in thalamic ECs at both P6 and P12 (Fig. 4b). Spatial validation using Visium, ISH, and Allen Brain Atlas datasets showed that *Tgfb2* is highly expressed in P6–P12 thalamus, whereas *Vegfa* is broadly expressed across regions and ages (Fig. 4c and Supplementary Fig. 6b–e). Our data indicate that *Tgfb2*–*Tgfb1*/*Tgfb2* interaction pairs accumulate the highest NicheNet scores, while other receptors (*Tgfb3*, *Eng*, *Acr1*) contribute minimally (Fig. 4a and Supplementary Data 3). Because *Tgfb1/2* are similarly expressed in CTX and THAL ECs, the thalamus-specific signaling arises from ligand enrichment (Supplementary Fig. 6f).

To investigate the cellular origin of ligand expression in our spatial datasets, we performed deconvolution of neuronal and glial signatures using the Zeisel et al. single-cell transcriptomic dataset²⁷. This analysis confirmed that most predicted NVU ligands originate from neurons, astrocytes, and oligodendrocytes and validated our Visium regional annotations, as inferred CTX- and THAL-specific cell-type distributions matched expected anatomical patterns (Fig. 4d, e and Supplementary Fig. 6g). Notably, *Tgfb2* expression mapped to *Rora*⁺ excitatory thalamic neurons (Fig. 4e and Supplementary Fig. 7a–c). In-situ hybridization on P12 sections confirmed *Tgfb2* mRNA colocalization with *Rora*⁺ thalamocortical neurons (Fig. 4f and Supplementary Fig. 7d–f). We further found that thalamic neurons express low levels of *Tgfb1* and *Tgfb3* (Fig. 4c and Supplementary Fig. 7b, c).

Endothelial cells expressed *Tgfb* ligands uniformly across regions and ages (Fig. 4f and Supplementary Fig. 6f), indicating that ECs are not the source of regionalized TGF β signaling, and excluding a cell-autonomous mechanism for the spatiotemporal heterogeneity observed. While astrocytic contribution cannot be entirely excluded with a low but detectable expression of *Tgfb2* (Supplementary Fig. 7b), our data strongly support a neuron-driven origin in the THAL region.

Together, these data indicate that neuronal, astrocytic, and oligodendrocyte maturation coincides with a vascular expansion in the thalamus and cortex, suggesting a coordinated neuro–glial–vascular remodeling program during this developmental window.

Endothelial TGF β signaling coordinates postnatal angiogenesis and vascular integrity mainly in the thalamic area

Given the thalamic enrichment of *Tgfb2* and the enhanced angiogenesis observed at P12, we tested whether endothelial TGF β signaling is required in a region- and time-specific manner using inducible endothelial *Tgfb1* knockout mice (*Tgfb1iECKO*). Tamoxifen induction at birth revealed no weight difference at P6, whereas mutants were 1.48-fold smaller than controls at P12 (Supplementary Fig. 8a). Although the phenotype was mild at P6, P12 *Tgfb1iECKO* mice displayed severe intracerebral hemorrhages, primarily within the thalamus (Fig. 5a and Supplementary Fig. 8b, e). Hemorrhages were restricted to the brain and retina, indicating a CNS-specific requirement for endothelial TGF β signaling (Supplementary Fig. 8c). Bleeding resolved by P20 without affecting survival (Supplementary Fig. 8d, e). Moreover, late postnatal deletion (P10–12 or P20–22) produced only mild or no hemorrhage, respectively, indicating that TGF β signaling is predominantly required during an early postnatal phase of thalamic angiogenesis (Fig. 5b, c and Supplementary Fig. 8f, g). CD31 and TER119 staining confirmed red blood cell extravasation (Fig. 5d). Mutants displayed enlarged, tortuous capillaries and ICAM2⁺ sacular malformations filled or surrounded by erythrocytes (Fig. 5e, f and Supplementary Fig. 8h, i).

At P12, mutants showed significantly more malformations in the thalamus than in the cortex (Fig. 5e, g), and cortical malformations, when present, were markedly smaller (Fig. 5h). Thalamic branching points were reduced 2.46-fold, with no change in the cortex (Fig. 5i).

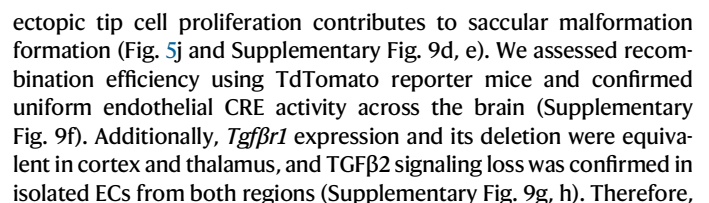


Fig. 4 | Identification of Region- And Age-specific NVU communications. **a** Left: Circos plot representation of enriched ligand–receptor interactions in each brain region at the corresponding age. The thickness of each connecting line is proportional to the ligand–receptor interaction weight derived from NicheNet’s signaling-weighted LR matrix (lr_sig), which reflects curated biochemical evidence and downstream signaling connectivity. Line transparency is also scaled to this weight, with more opaque lines indicating stronger predicted interactions. Right: Venn diagram using ligands from P6, P12, and adult brain regions. Potential angiogenic ligands common in the P6 and P12 regions are listed. **b** Ligand activity scores for predicted region- and age-enriched interactors. **c** Visium plot of *Tgfb1*, *Tgfb2*, and *Tgfb3* expression in P6, P12, and adult brain regions. **d** Visium plot showing deconvoluted thalamic cell-type identity based on RCTD analysis using brain single-cell RNAseq data from Zeisel et al. Cell, 2018. **e** Dot plot displaying

THAL-enriched ligands expression across neuronal and glial populations using the Zeisel et al. single-cell reference dataset. **f** Left, representative image of triple *Tgfb2*, *Tgfb3* and *Rora* mRNA labeling in P12 thalamic area. Nuclei were labeled by DAPI. Bottom, high magnification picture of the white dashed square. The yellow dashed lines delimited two *Rora*+ nuclei expressing *Tgfb2* (red) and *Tgfb3* (green) mRNA. Right, quantifications of the number of *Tgfb2* mRNA dots in microglial cells (IBA1+), endothelial cells (ERG1/2/3+), pericytes (PDGFRβ+) and thalamocortical neurons (*Rora*+). The bold number in front of each column represents the total number of cells quantified in the thalamic area of 4 P12 mice. Values represent means ± s.e.m, *P* values (IBA1+ vs. ERG+ *p* = 0.0435, IBA1+ vs. *Rora* + *p* < 0.0001, ERG+ vs. *Rora* + *p* < 0.0001, PDGFRβ+ vs. *Rora* + *p* < 0.0001) were calculated by two-sided Kruskal-Wallis test, Dunn’s multiple comparisons test (**f**). (CTX: cortex; THAL: thalamus). Source data are provided as a Source Data file.

the spatiotemporal vascular phenotype observed in the *Tgfb1iECKO* mice validates our findings on NVU communications.

mTORC1 inhibition prevents vascular anomalies in *Tgfb1iECKO* mice

To investigate mechanisms downstream of endothelial TGFβ signaling, we performed scRNAseq on ECs from *Tgfb1iECKO* and control mice. UMAP analysis and subcluster annotation (Fig. 6a and Supplementary Fig. 10a–c) confirmed a broad reduction of TGFβ signaling signatures in all mutant clusters (Supplementary Fig. 10d). Mutants displayed a 2.01-fold decrease in capillaries and a 1.82-fold increase in venous ECs (Supplementary Fig. 10b). Although the proliferative EC population was unchanged, most mutant clusters, including tip cells, showed increased S/G2/M and reduced G1 signatures (Supplementary Fig. 10e), consistent with the enhanced proliferation and capillary rarefaction observed in Fig. 5. UMAP representations indicated that capillary and tip cells were the most transcriptionally affected (Fig. 6a). Mutant ECs showed loss of D-tip markers such as *Apod* and defects in BBB-, WNT-, WNT-target, and ECM-related programs, including reduced *Tfrc*, *Mfsd2a*, and *Spock2* and increased *Cav1* (Fig. 6b, c; Supplementary Fig. 10f, g and Supplementary Fig. 11a, b). These permeability-associated defects likely contribute to hemorrhage. Intriguingly, pericyte coverage remained unaffected (Supplementary Fig. 11c).

Pathway analysis revealed mTORC1 enrichment in capillary, proliferative, and tip cell clusters (Fig. 6d and Supplementary Data 4). Immunostaining confirmed strong phospho-S6 and phospho-4EBP1 in malformations and mutant capillaries, with no signal in controls (Fig. 6e and Supplementary Fig. 12a). A milder increase was detected in cortical tip cells and in CTX amino acid metabolism, but both effects were more pronounced in the thalamus (Supplementary Fig. 12b, c). In vitro, TGFβ2 suppressed VEGFA/VEGFC-induced mTOR activation without affecting AKT (Supplementary Fig. 12d–f), supporting a direct TGFβ2–TGFβR1–mTOR regulatory axis.

We next tested whether mTORC1 inhibition could rescue the phenotype. *Tgfb1iECKO* mice treated with rapamycin (2 mg/kg, P6/P9) exhibited a marked reduction in thalamic hemorrhage (Fig. 6f) and suppressed endothelial S6 phosphorylation, along with strong decreases in malformation number (4.26-fold) and malformation area (2.48-fold). Capillary branchpoints improved 1.50-fold, and tip cell formation showed a strong trend toward recovery (10.13-fold, *p* = 0.0542) (Fig. 6f and Supplementary Fig. 13a, b). Rapamycin also restored tip-cell morphology (Supplementary Fig. 13c). To confirm endothelial specificity, we generated *Rptor*; *Tgfb1iECKO* double mutants. Endothelial *Rptor* loss reduced S6 phosphorylation, decreased malformations and bleeding, and tended to increase tip-cell number (7.95-fold, *p* = 0.0687), mimicking rapamycin treatment (Fig. 6g and Supplementary Fig. 13d). Collectively, these results indicate that thalamic malformations and hemorrhages arise from BBB/ WNT/ECM defects and abnormal cell-cycle activity linked to mTORC1 hyperactivation in *Tgfb1*-deficient capillaries and tip cells.

Discussion

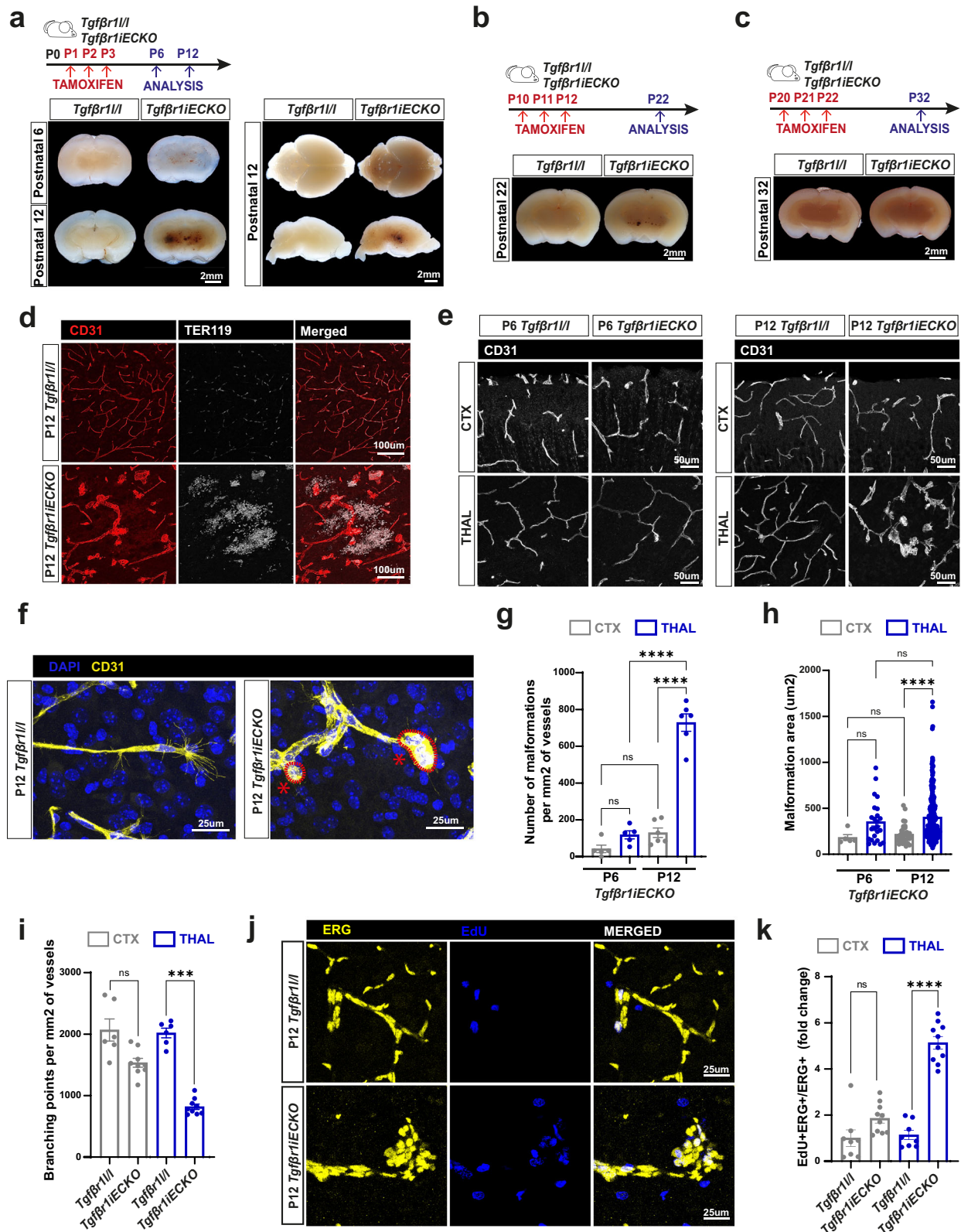
NVU communications are essential for vascular patterning, BBB function, and cerebral blood flow, thereby maintaining neuronal homeostasis. While neuronal activity-dependent angiogenesis has been well characterized in the cortex, how region-specific cellular heterogeneity shapes angiogenic mechanisms across the developing brain remains unclear. This study aimed to address this gap by combining spatial transcriptomics and endothelial single-cell RNA sequencing in the postnatal mouse brain to map the molecular cues of developing brain regions and generate a spatiotemporal framework of NVU communication.

We first observed regional vascular heterogeneity emerging by P6 and persisting into adulthood, consistent with whole-brain vascular reconstructions^{7,8}. We observed similar postnatal angiogenic waves consistent with previous findings^{11,17}, and higher tip-cell abundance in the thalamus suggests region-specific remodeling mechanisms contributing to its elevated vascular density. scRNAseq showed that cortical and thalamic endothelial cells diverge transcriptionally as early as P6, with temporal changes outweighing regional ones and revealing distinct sprouting, migratory, and BBB-associated signatures across P6, P12, and adulthood. It is also established that adult mice display region-specific levels of BBB permeability^{11,12}, supporting the concept of sustained vascular heterogeneity.

Spatial transcriptomics resolved region-specific transcriptomic programs across 20,000 genes, highlighting coordinated transitions in synaptic remodeling, glial differentiation, and neuronal maturation. These findings align with those of Thompson et al., who demonstrated spatiotemporal regulation of developmental genes using in situ hybridization²⁸.

NVU communication analysis identified both shared and region-enriched ligand–receptor interactions. *Vegfa* remained uniformly high across regions and ages, consistent with roles in vascular homeostasis²⁹, supporting neuronal and glial survival^{30–33}, and modulating recovery in ischemia^{34,35}. Furthermore, increasing VEGFA signaling improves organ function, prevents capillary loss, and extends lifespan in aged mice³⁶, underscoring its broader importance. The need for more selective angiogenic regulators is therefore evident, particularly those affecting sprouting and remodeling.

Region-resolved markers aligned with expected anatomical patterns, underscoring how neuronal and glial remodeling progress in parallel with vascular development. NVU communication signals evolve from P6 to adulthood, aligning with the neuronal, astrocytic, and oligodendrocyte maturation signatures revealed by spatial transcriptomics. Previous studies by Lacoste et al.¹⁵ and Whiteus et al.²⁰ have shown that altering neuronal activity during this developmental window reshapes cortical vascularization, highlighting the functional sensitivity of this period. Importantly, the developmental increase in neural and glial maturation between P6 and P12 paralleled the vascular remodeling dynamics described in Fig. 1, namely the marked decline in



endothelial proliferation after P12 and the sustained presence of tip cells. These parallel trajectories suggest that neural–glial maturation actively shapes regional endothelial remodeling rather than simply coinciding with it.

The NVU communication map provides a conceptual framework for understanding regional determinants of neurovascular vulnerability. Vascular dysfunction contributes to small vessel disease (SVD),

Alzheimer’s disease, stroke, glioma, autoimmune disorders, and neurodegenerative diseases^{37–39}. Despite significant efforts to develop angiogenic therapies for brain disorders^{34,40}, their efficacy remains elusive due to an incomplete understanding of regional NVU communications. Understanding region-specific cues and their developmental dynamics may clarify why certain regions show selective vulnerability.

Fig. 5 | Endothelial TGF β signaling coordinates postnatal angiogenesis and vascular stabilization mainly in the thalamic area. **a** Top, schematic representation of experimental strategies to induce postnatal endothelial deletion of *Tgfb β 1* gene. Bottom, representative visual images of P6 and P12 brain coronal sections (left) and P12 brain sagittal sections (right bottom) from control and mutant mice injected at birth. **b, c** Top, schematic representation of experimental strategies to induce postnatal endothelial deletion of *Tgfb β 1* gene. Bottom, brain coronal sections of P22 (**b**) and P32 (**c**) control and mutant mice injected at P10 and P20, respectively. **d** Double CD31 and TER119 immunolabeling of the thalamic area of P12 control and mutant mice. **e** CD31 immunolabeling of the cortical and thalamic area from P6 and P12 control or mutant mice injected at birth. **f** High magnification pictures of growing capillary tip in the thalamic area from P12 control or mutant mice using CD31 immunolabeling and DAPI labeling. Malformations are

indicated with red asterisks and outlined with red dashed lines at the edges of CD31+ capillaries. **g–i** Quantifications of the vascular malformation number per mm² of vessels (**g**), the area (μ m²) of the malformations (**h**) and the branching points per mm² of vessels (*Tgfb β 1*/I THAL vs. *Tgfb β 1*IECKO THAL $p = 0.0002$) (**i**). **j, k** Double ERG and EdU immunolabeling of proliferative ECs in the thalamic area of P12 brains (**j**) and corresponding quantification in the cortical and thalamic area of P12 control or mutant mice (**k**). Dot plots represent pictures from at least 4 brains per condition (**g, h, i, k**). Values represent means $\pm$ s.e.m, P values (*** $P < 0.001$, **** $P < 0.0001$) were calculated by ordinary one-way ANOVA with Tukey's multiple comparisons test (**g, k**) and Kruskal-Wallis with Dunn's multiple comparisons test (**h, i**). All statistical tests were two-sided. (CTX: cortex; THAL: thalamus). Source data are provided as a Source Data file.

Several region-enriched NVU ligands identified during early postnatal development correspond to disease processes known to originate from developmental disruptions in those same brain regions. Thalamic ligands such as FGF9 are implicated in disorders involving thalamocortical developmental abnormalities, including major depressive disorder⁴¹ and MS-related neurodegeneration⁴². S100B, also enriched in the thalamus, contributes to early neuroinflammatory processes underlying epilepsy and neurodevelopmental injury^{43,44}. In contrast, cortex-enriched signals such as RELN and SLITRK5 regulate cortical circuit formation and are disrupted in autism and psychiatric phenotypes^{45,46} and OCD-related behaviors⁴⁷. These disorders all involve neurovascular dysfunction, and our NVU communication map provides a framework to explore how early region-specific endothelial–neuronal interactions may contribute to such vulnerabilities.

We also identified neuronal *Tgfb β 2* as highly expressed in the THAL at P6–P12, driving selective activation of TGF β signaling in thalamic ECs. Endothelial *Tgfb β 1* deletion induced vascular malformations and hemorrhages predominantly in the THAL within this same developmental window, validating our NVU communication map. We found that postnatal brain vascular remodeling relies on specialized neuronal D-tip cells that we previously described in the neuroretina²², and loss of TGF β signaling caused reduced D-tip identity, increased proliferation, and BBB/WNT/ECM defects. These defects align with embryonic TGF β -related vascular phenotypes^{26,48–50}. Together with these phenotypes, the developmental context provides a coherent explanation for the role of thalamic TGF β 2 during the P6–P12 window: as neural circuits mature, TGF β normally restrains endothelial proliferation and stabilizes tip-cell–driven remodeling, meaning that loss of TGF β 1 signaling at this stage removes an important brake on angiogenic activity.

Mechanistically, loss of TGF β signaling increased growth factor–induced mTOR activation in vitro and in vivo, identifying TGF β 2/ALK5 as a broad inhibitor of mTOR signaling, as previously shown in immune and smooth muscle cells^{31–33}. mTOR inhibition rescued hemorrhages, reduced malformation number and size, decreased S6 phosphorylation, and partially restored branching, confirming mTOR hyperactivation as a major driver of the pathology. Our findings are consistent with previous findings showing that mTOR inhibition protects BBB integrity^{54,55}. However, the precise mechanisms by which TGF β signaling regulates growth factor–induced mTOR activity and EC proliferation remain unclear and require further investigation. This regulatory relationship is particularly relevant in the context of TGF β -related vascular disorders such as Loews–Dietz syndrome (LDS)^{56,57} and CARASIL⁵⁸, where defective TGF β signaling underlies vascular instability. The overlap suggests that mTOR pathway modulation may offer therapeutic potential in settings of TGF β signaling insufficiency.

It is crucial to consider the role of the microenvironment in angiogenic processes and BBB maintenance, as described for the

induction of BBB properties in leaky vessels of the circumventricular organs by the WNT signaling pathway^{59,60}. Furthermore, it is essential to acknowledge that our NVU communication map is built on an evolving and expanding database, and the validation of region-enriched cues is challenging and requires further investigations on other brain regions, primarily due to the recent emergence of the concept of cerebral vascular heterogeneity. As new data becomes available, the NVU communication map will continue to be refined and updated, ensuring its relevance and accuracy.

Together, our integrated spatial and single-cell analyses reveal that neuronal and glial maturation programs drive region-specific endothelial states and angiogenic trajectories during a critical postnatal window. By mapping these NVU communication dynamics, we define the developmental architecture of brain vascular heterogeneity and provide a framework for interpreting region-selective neurovascular vulnerability.

Methods

Mice models and experimental protocols

Tgfb β 1 postnatal endothelial-specific deletion was generated by crossing *Tgfb β 1*/I mice (mixed background, #028701, Jackson Laboratory)²² with *Cdh5CreERT2* mice⁶¹. The *Tgfb β 1*/I *Tdtl*+/ *Cdh5CreERT2* mice were generated by crossing *Tgfb β 1*/I *cdh5CreERT2* mice with TdTomato reporter mice B6.Cg-Gt(ROSA)26Sortm14(CAG-tTomato)Hze/J (#007914, Jackson Laboratory). The double mutant *Rptor*/I *Tgfb β 1*/I *Cdh5CreERT2* mice were generated by crossing *Rptor*/I mice (#013188, Jackson Laboratory) with *Tgfb β 1*/I *Cdh5CreERT2* mice. The deletion of the genes was induced by intraperitoneal injections of 100 μ g of tamoxifen (Sigma, T5648; stock solution at 10 mg/mL in 10% ethanol and 90% corn oil and injected at 2 mg/mL in corn oil) at P1/P2/P3 for the early postnatal deletion and P10/P11/P12 or P20/P21/P22 for the late postnatal deletions. Cre-negative littermates were treated with tamoxifen and used as control mice for each experiment. Rapamycin (Selleckchem, S1039; stock solution at 100 mg/mL in DMSO) was injected intraperitoneally (2 mg/kg in corn oil) at P6 and P9. For the study of postnatal angiogenesis, C57BL/6j mice (#000664, Jackson Laboratory) were used from P1 to P40. For the in vivo proliferation assay, mice were intraperitoneally injected with 30 mg/kg of EdU (Life Technologies Inc., A10044, stock solution at 5 mg/mL). The Comité Institutionnel des Bonnes Pratiques Animales en Recherche (CIBPAR) approved all the breeding colonies and experimental protocols (#2024-6395; #2024-6396). Mice were housed in a pathogen-free facility under a 12 h light/12 h dark cycle at 22 $\pm$ 2 °C with 40–60% humidity and ad libitum access to food and water. Pups younger than P21 were subsequently decapitated, whereas older animals underwent cervical dislocation prior to tissue collection. Unless otherwise specified, both male and female mice were used. For spatial and single-cell transcriptomic experiments using C57BL/6j mice, only males were used. Sex was not recorded in certain early postnatal experiments, in which litters were pooled prior to analysis.

Fig. 6 | mTORC1 inhibition rescues vascular anomalies in *Tgfb β 1iECKO* mice. **a** UMAP plot of brain EC subclusters from P12 control (*Tgfb β 1i/l*, left) and mutant (*Tgfb β 1iECKO*, right) mice. **b** Volcano plot of differentially expressed genes between P12 mutant and control capillary (left) or tip (right) cells. Genes that are significantly differentially expressed ($p < 0.05$, p -value were calculated by the Wilcoxon Rank Sum test). For aesthetic reasons, only certain genes of interest are labeled. **c** Double CD31 and CAV1 immunolabeling of the thalamic area from P12 control or mutant mice (representative image of staining observed in at least 3 mice per group). **d** Heatmap of the mTOR signaling pathway activity in *Tgfb β 1iECKO* compared to control, with their respective p -value calculated by the Wilcoxon Rank Sum test. **e** Double CD31 and p -S6 immunolabeling of the thalamic area from P12 control or mutant mice (representative image of staining observed in at least 3 mice per group). **f, g** Top, schematic representation of experimental strategies to inhibit

mTOR with rapamycin injection. Middle, Representative visual images of brain coronal sections from P12 mutant mice treated with DMSO or 2 mg/kg of rapamycin (**f**), or with the indicated genotype (**g**). Bottom, quantifications of vascular malformations per mm² of vessels and the number of tip cells per mm² of vessels in the thalamic area from mutant P12 *Tgfb β 1iECKO* mice treated with DMSO or rapamycin (*Tgfb β 1iECKO* DMSO vs. *Tgfb β 1iECKO* RAPAMYCIN $p = 0.0003$) (**f**) or from P12 *Rptorl/+; Tgfb β 1iECKO* and *Rptorl/l; Tgfb β 1iECKO* mice (*Rptl/+ Tgfb β 1iECKO* vs. *Rptl/+ Tgfb β 1iECKO* $p = 0.0378$) (**g**). Each dot is representative of one mouse (**f, g**). Values represent means $\pm$ s.e.m, P values ($*P < 0.05$, $***P < 0.001$) were calculated by two-sided Welch's t-test, two-sided unpaired t-test or two-sided Mann-Whitney test (**f, g**). (UMAP: Uniform Manifold Approximation and Projection). Source data are provided as a Source Data file.

blocking buffer at RT (Supplementary Table 1). For P40 mice, immunofluorescence protocol was optimized from P1 to P20 mice to achieve similar staining quality, brains sections were permeabilized 3 h with 20% DMSO (Sigma Aldrich, 276855), 5,75% glycine (Sigma Aldrich, G7126), 0,2% Triton X-100 (Sigma Aldrich, X100) in PBS at 37 °C and blocked O.N. at 37 °C with blocking solution (0,2% of gelatin (Sigma Aldrich, G1393), 0,2% Triton X-100 (Sigma Aldrich, X100), 0,01% sodium azide (Sigma Aldrich, S2002) in PBS). Then, brain sections were incubated with primary antibody for 2 days, followed by O.N secondary antibody in blocking solution at 37 °C. After PBS washes, brain slices were incubated for 5 min with 2 μ g/mL DAPI (Sigma, D9542, stock solution at 10 mg/mL in H₂O) and mounted with fluorescent mounting medium (Agilent, S302380-2).

For the analysis of the vascular malformations in *Tgfb β 1iECKO* mice, *Rptorl/+; Tgfb β 1iECKO* and *Rptorl/l; Tgfb β 1iECKO* mice, the vascular malformations were defined as saccular malformations at the edge of CD31+ capillaries. The number and the area were quantified using Fiji software by delineating the boundaries of the saccular malformations.

For the in vivo proliferation assay, click it EdU Cell Proliferation kit (Invitrogen, C10340) was used according to the manufacturer's instructions using 30 μ m thick brain sections of mice previously injected with EdU.

For in situ hybridization, 30 μ m PFA-fixed floating sections were used according to the manufacturer's instructions of RNAscope LS Multiplex Fluorescent v2 Assay for fixed-frozen tissue samples (Advanced Cell Diagnostics). Brain sections were washed twice for 10 min in PBS, placed on Superfrost Plus microscope slides (Fisher Scientific, 12-550-15), and air-dried for 5-10 min at room temperature. Slides with tissue sections were baked 30 min at 60 °C, and fixed for 15 min in 4% PFA. Then tissues were dehydrated with croissant ethanol solutions (50%, 70%, and 100%), air dried for 5 min, and dehydrated slides were incubated with hydrogen peroxide for 10 min. After incubation with protease IV at RT for 30 min and two washes with PBS, probes (Supplementary Table 1) were hybridized for 2 h at 40 °C and twice washed with wash buffer. For the amplification reaction, slides were incubated sequentially with RNAscope Multiplex FL v2 Amp 1, then with RNAscope Multiplex FL v2 Amp 2 for 30min at 40 °C each, and finally with RNAscope Multiplex FL v2 Amp 3 for 15min at 40 °C. Slides were washed at each step with the wash buffer. Then, depending on each probe, slides were incubated with the appropriate RNAscope Multiplex FL v2 HRP-C1, 2 or 3 for 15min at 40 °C, washed twice with Wash Buffer for 2min at RT, then incubated with the appropriate fluorophore and RNAscope Multiplex FL v2 HRP blocker for 15min at 40 °C. After washing the slides for 2min at RT, the slides were immunofluorescence-stained with ERG1, 2, and 3, PDGFR β , or IBA1 as needed, incubated with DAPI, and mounted as previously described. Brain slides were imaged with a Sp8 confocal at higher magnification 63x (Leica TCS Sp8 laser scanning confocal microscope), and the punctate dots in the nuclei of the cells of interest were quantified with Fiji software.

Analysis of postnatal brain angiogenesis

Vessel segmentation. Blood vessels were stained with antibodies against CD31 and ICAM2. Brain slides were imaged with a Leica DMI8 inverted microscope, and both CD31 and ICAM2 pictures were merged to improve the brain vasculature's signal-to-noise ratio. Then using Imaris 9 software (Imaris 9.9.1) we provided a segmented picture of blood vessels for each postnatal time point. Briefly, "Surfaces" module was used on CD31/ICAM2 merged channels, and the surfaces detail was set at 1 μ m and background subtraction at 3 μ m. Pixel classification was used to train and predict false-positive vessels.

The analysis of the segmented data for the Heatmap creation and the analysis of vessel density per brain area were performed using Python and Scikit-Image (S. Van der Walt et al., 'scikit-image: image processing in Python', PeerJ, vol. 2, p. e453, 2014). All the code used for analysis is published in the 'LIOM Toolkit' Python package, specifically the segmentation module (F. Irgolitsch, 'LIOM Toolkit', GitHub repository. GitHub, 2023. [Online]. Available:<https://github.com/LIOMLab/liom-toolkit>).

Heatmap creation. To visualize the brain vascular density, heatmaps were created from the blood vessel segmentation pictures described above. The segmented pictures were divided up into squares of fixed height and width, to ensure every square has the same area at all time points. This is possible due to all images having the same voxel size. The number of pixels with blood vessels was summed in each square to set the intensity of the heatmap. To ensure proper scaling across images with different maximum intensities, the final square in the bottom right was set to maximum intensity to ensure each image has the same maximum intensity to keep the calibration bar consistent across all images. This square was outside the sample region in every image. Then, combined with the fixed square size, we could compare heatmaps of different brain sections across different time points.

Analysis of the vessel density per brain area. Brain regions from the acquired slices were delineated based on the Allen brain reference atlas, and masks were created using Adobe Illustrator. These masks were subsequently applied to the segmented images to quantify the vessel area specific to each brain region. The regional area was determined by calculating the area of each corresponding mask. Vascular density was then computed by dividing the vessel area by the region's area. To obtain the physical dimensions of the region and vessel areas, the masks were multiplied by their respective voxel size. Finally, the vascular density values were normalized by the regional area to express the results as a percentage of vessel density within each brain region.

Spatial transcriptomics

Tissue preparation, RNA amplification, and library preparation. Brains ($n = 2$ males per age; P6, P12 and adult; C57BL/6J strain) were simultaneously frozen and embedded in OCT for cryo-sectioning at 10 μ m thickness (Allen brain reference atlas, developing mouse brain

coronal, corresponding to section 236 at P7, 255 at P14, and 75 at P56, (<https://developingmouse.brain-map.org/experiment/siv?id=100098039&imageId=101617160&initImage=nissl>, <https://developingmouse.brain-map.org/experiment/siv?id=100014118&imageId=100500200&initImage=nissl>). Reactions were carried out with the Visium Spatial Gene Expression (GEX) and Tissue Optimization (TO) Slide & Reagent Kits according to the manufacturer's protocol with recommended reagents (10X Genomics, 1000193 and 1000187). Brain sections were placed on designated capture areas of slides for TO and GEX and stored at -80°C until further processing. TO and GEX slides were fixed in methanol solution and stained with hematoxylin and eosin (H&E) for visualization of tissue morphology on an automated slide scanner (Zeiss Axioscan Z1). Using TO slides, we selected 30 min as the optimal permeabilization time. GEX slides were incubated with RT, and cDNA was amplified for 15 cycles. Library construction steps were performed according to the manufacturer's protocol and included cDNA fragmentation, end repair and A-tailing, adaptor ligation, and sample indexing and amplification. Quality control of the constructed library was conducted via Bioanalyzer (Agilent). Illumina sequencing of the resulting libraries was performed by Genome Quebec on an Illumina NovaSeq S4 (Illumina). Base calling, demultiplexing, and generation of FastQ files were conducted by Genome Quebec.

Data processing. Gene expression matrices were generated using SpaceRanger v1.3.1 (10X Genomics) using the mm10-2020-A as reference genome for P6 (Replicate 1: Slide V12U06-099-D1, mm10-2020-A, Spatial 3' v1; Replicate 2: Slide V20N02-081-A1, mm10-2020-A, Spatial 3' v1), P12 (Replicate 1: Slide V12U06-099-C1, mm10-2020-A, Spatial 3' v1; Replicate 2: Slide V20N02-081-D1, mm10-2020-A, Spatial 3' v1) and adult (Replicate 1: Slide V12U06-099-B1, mm10-2020-A, Spatial 3' v1; Replicate 2: Slide V12U06-099-A1, mm10-2020-A, Spatial 3' v1) coronal brain slices (Supplementary Fig. 3a). Further processing and analysis of the data was performed using R v4.4.0. The Seurat v5.1.0 R-package⁶² was used to analyze the filtered gene expression matrix output from SpaceRanger. All spots and genes from the filtered expression matrix were retained in the analysis. To account for technical artifacts, while preserving biological variance, the spatial gene expression matrices for each sample were normalized using the SCTransform function⁶³. Principal components analysis was conducted on the normalized expression matrix using the RunPCA function. To correspond to the Allen atlas mapping, a nearest neighbor search was conducted on the first 40 PCA embeddings on individual P6, P12 and Adult brain sections, respectively, using the FindNeighbors function with all other parameters set to their default values and the resulting spatial clustering was visualized using the SpatialDimPlot function. Spots were clustered from the resulting nearest neighbors graph using the FindClusters function. To visualize spatial transcriptomic data in latent space, dimensional reduction of the same PCA embeddings resulting from the nearest neighbor search was conducted using Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP)⁶⁴. Significant (adjusted p -value < 0.05) marker genes for each cluster were identified by running the FindAllMarkers function on the normalized expression matrices. UMAP-derived clusters were annotated based on their marker genes and spatial location.

To elucidate spatiotemporal changes in gene expression, Seurat objects of each brain slices were integrated into a single object (IntegrateLayers using CCAIntegration on re-calculated SCT values) followed by dimensionality reduction using UMAP on the first 35 PCs then clustering based on UMAP embeddings. Region annotations from the individual Visium dataset were used as a reference to reannotate the integrated dataset. Pathway activity score was calculated with the VISION package⁶⁵ on SCT data using the msigdb.v2023.1.Mm pathway database^{22,66}, then added to the Seurat object as a new assay (VISION). Differential gene expression analysis between each annotated region-

timepoint pair was conducted using the FindMarkers function with default parameters on SCT data (gene expression) or VISION data (Pathway expression). Pseudobulk analysis using the Pseudobulk-Expression function was performed on either normalized or normalized and scaled data. SpatialFeaturePlot was used to visualize spatially the expression of genes (SCT) or pathways (VISION) while DotPlot was used to assessed gene expression level (Average and % of expression) per regions. Deconvolution of spatial count data was performed using RCTD from the spacexr package (<https://github.com/dmccable/spacexr>) in full mode using cell-specific RNA counts from selected regions at P19-21 of the Zeisel et al. scRNAseq dataset as reference²⁷, and weight-normalized results were visualized with SpatialFeaturePlot.

Single-cell RNA sequencing

Sample preparation, immunostaining, and gating strategy of the C57BL/6J strain. Brains were placed in a chilled brain slicer (WPI; rodent brain matrices acrylic) to obtain 1 mm thickness of coronal brain sections. According to Allen brain atlas reference, isocortex and thalamic area were microdissected using microscalpel (WPI, 500250) under microscope in ice cold dissociation medium solution (DM; Hank's balanced salt solution without calcium and magnesium (ThermoFisher, 14175095), 10 mM HEPES (ThermoFisher, 15630080), 9 mM MgCl₂ (Fisher Scientific, AAJ61014EQE), 35 mM D-glucose (Fisher Scientific, AAJ60067AK), pH=7.35). After slicing into little pieces, brains were enzymatically digested using Neural Tissue Dissociation Kit (P) (Miltenyi Biotec, 130-092-628) for 15 min in mix 1 and 10 min in mix 2 at 33°C . During the digestion process, tissues were homogenized two times by up and down pipetting. At the end of the digestion process, samples were put in the ice with DM solution and cell clusters were removed by filtration through prewetted $70\text{ }\mu\text{m}$ and centrifuged at $200 \times g$ for 8 min at 4°C . To stop the enzymatic reactions, the pellet is resuspended in DM solution with 3 mg of ovomucoid protease (Worthington, LK003182, resuspended in DM solution) and 300 units of DNaseI (Worthington, LK003172, resuspended in DM solution). The myelin and the cell debris were removed by using the debris removal solution (Miltenyi Biotec, 130-109-398). After 10 min blocking at RT with $5\text{ }\mu\text{g/mL}$ of CD16/CD32 Fc Block (eBiosciences, 16-0161-82), the cells were stained at RT for 20 min with APC anti-mouse CD31 (BD Biosciences, 551262) and live/dead fixable Aqua dead (ThermoFisher Scientific, L34957), allowing the staining of dead cells. After the staining process, the cell pellet is fixed between 16 h and 20 hours at 4°C according to the manufacturer's instructions (10X Genomics, Chromium Next GEM Single Cell Fixed RNA Sample Preparation Kit). After a washing step, the stained cells were resuspended with 2 mM of EDTA and 0.04 % of BSA. The setting up of the gates and compensations was allowed with the unstained cells, the single positive staining, the Fluorescence Minus One (FMO) and the isotype control staining (APC Rat IgG2a, κ (BD Biosciences, 551139). After removing the dead cells and doublets, endothelial cells (CD31 +) were sorted using a BD FACS Aria Fusion from 26 P6, 19 P12 and 39 2-month-old mice and stored at -80°C in buffer.

Gating strategy of C57BL/6J strain. After slicing into little pieces, brains were enzymatically digested using Neural Tissue Dissociation Kit (P) (Miltenyi Biotec, 130-092-628) for 15 min in mix 1 and 10 min in mix 2 at 33°C . During the digestion process, tissues were homogenized two times by up and down pipetting. At the end of the digestion process, samples were put in the ice with DM solution and cell clusters were removed by filtration through prewetted $70\text{ }\mu\text{m}$ and centrifuged at $200 \times g$ for 8 min at 4°C . To stop the enzymatic reactions, the pellet is resuspended in DM solution with 3 mg of ovomucoid protease (Worthington, LK003182, resuspended in DM solution) and 300 units of DNaseI (Worthington, LK003172, resuspended in DM solution). The myelin and the cell debris were removed by using the

debris removal solution (Miltenyi Biotec, 130-109-398). After 10 min blocking at RT with 5 µg/mL of CD16/CD32 Fc Block (eBiosciences, 16-0161-82), the cells were stained at RT for 20 min with APC anti-mouse CD31 (BD Biosciences, 551262) and live/dead fixable Aqua dead (ThermoFisher Scientific, L34957), allowing the staining of dead cells. After the staining process, the cell pellet is fixed between 16 h and 20 hours at 4 °C according to the manufacturer's instructions (10x Genomics, Chromium Next GEM Single Cell Fixed RNA Sample Preparation Kit). After a washing step, the stained cells were resuspended with 2 mM of EDTA and 0.04 % of BSA. The setting up of the gates and compensations was allowed with the unstained cells, the single positive staining, the Fluorescence Minus One (FMO) and the isotype control staining (APC Rat IgG2a, κ (BD Biosciences, 551139). After removing the dead cells and doublets, endothelial cells (CD31+) were sorted using a BD FACS Aria Fusion from 26 P6, 19 P12 and 39 2-month-old mice and stored at -80 °C in buffer.

Gating strategy of *Tgfr1*l*EKO* strain. Four brains of P12 *Tgfr1*l*EKO* and littermate control (*Tgfr1* l/l) (*n* = 4 brains per genotype) were enzymatically digested using Neural Tissue Dissociation Kit (P) (Miltenyi Biotec, 130-092-628) for 30 min at 37 °C. During the digestion process, tissues were homogenized two times by up and down pipetting. The samples were put in the ice with 5% SVF DMEM to stop the enzymatic reaction and pass through a 40 µm cell strainer to remove cell debris (Falcon, 352340). After a PBS step washing, the myelin and the cell debris were separated from the pelleted cells by using the debris removal solution (Miltenyi Biotec, 130-109-398) and the cells were washed with PBS before proceeding to the red blood cell lysis (Miltenyi Biotec, 130-094-183). After 30 min blocking of 5 µg/mL of CD16/CD32 Fc Block (eBiosciences, 16-0161-82), the cells were stained at RT for 40 min with a combination of the following antibodies (PE anti-mouse CD140b Antibody (Biolegend, 136006); APC anti-mouse CD31 (BD Biosciences, 551262); FITC anti-mouse CD45 (Biolegend, 103108)). After washing the cells with PBS, the cells were stained 20 min at room temperature with live/dead fixable Aqua dead (ThermoFisher Scientific, L34957), allowing the staining of dead cells. After a washing step, the stained cells were resuspended with 2 mM of EDTA and 0.04 % of BSA. The setting up of the gates and compensations were allowed with the unstained cells, the single positive staining, the Fluorescence Minus One (FMO) and the isotype control staining (PE Rat IgG2a, κ Isotype control antibody (BD Biosciences, 557229); APC Rat IgG2a, κ (BD Biosciences, 551139) and FITC Rat IgG2b, κ (Biolegend, 400633)). After removing the dead cells and doublets, 10 000 endothelial cells (CD45- CD31+ CD140b-) were sorted with a BD FACS Aria Fusion for each sample.

Library preparation of C57BL/6 J strain single cell suspension. Single-cell suspensions of fixed and sorted brain cells from P6, P12 and adult Thalamus and Cortex regions (12 datasets in total) were resuspended in PBS containing 0.04% ultra-pure BSA. Multiplexed libraries were prepared using the Chromium Next GEM Single Cell Fixed RNA Kit, Mouse Transcriptome, 4rxns x 4 BC (# 1000496, 10x Genomics; Pleasanton, CA, USA) according to the manufacturer's instructions. The target recovery for each sample was 10,000 cells. Generated FLEX libraries were sequenced on an Illumina NovaSeq6000 S4, followed by de-multiplexing and mapping to the mouse transcriptome probe set (Chromium_Mouse_Transcriptome_Probe_Set_v1.1.1.GRCm39-2024-A) using CellRanger multi (pipeline version 9.0.1) (10x Genomics).

Data processing and EC clustering of the scRNAseq of C57BL/6 J strain. Further processing and analysis of the 12 scRNAseq datasets were performed using R (version 4.4.0) as previously described²². For each FACS sample, the filtered gene expression matrix was analyzed using tools from the Seurat R-package v5.1.0⁶². The endothelial-enriched sample (6 CD31+ datasets) were selected for further analysis

for additional quality control steps filtering out genes that were expressed in less than 3 cells and cells that expressed fewer than 100 genes. Lastly, only cells which had fewer than 30000 transcripts, fewer than 10000 genes, and less than 15% of transcripts originating from mitochondrial genes were retained for further analysis. All 6 datasets were merged and normalized gene expression data, generated using the SCT transform function (regressing on nFeature_RNA and percent.mt), was used for dimensionality reduction and clustering, while differential gene expression analysis was performed on log normalized RNA assay. After subclustering of EC clusters (based on EC markers) from P12 Control and P12 *Tgfr1*l*EKO* datasets, principal components analysis was conducted on the SCT normalized expression matrix using the RunPCA function. Dimensional reduction of the data was performed using Uniform Manifold Approximation and Projection (UMAP)⁶⁴ as implemented in the *RunUMAP* function, where the number of PCA dimensions used for the *dims* parameter was 15, and all other settings were default. Cells were clustered from the resulting UMAP dimensions using the FindClusters function with a resolution of 0.5 with all other parameters set to their default values. Cell clusters were annotated based on well-known marker gene expression, namely, EC clusters were identified by the presence of *Cldn5* and *Pecam1* expression while EC subtypes were defined using markers as previously published^{13,21,22}. AverageExpression function on time and location specific EC subtypes was used to performed PCA analysis on high variable features with the *prcomp* function (R stats package). Pathway activity score was calculated with the VISION package⁶⁵ on SCT data using the msigdb.v2023.1.Mm pathway database and a collection of expert annotated gene sets generated in-house⁶⁶, then added to the Seurat object as a new assay (VISION). Differential gene expression analysis between each annotated EC subtypes was conducted using the FindMarkers function on SCT assay with default parameters and DEGs were visualized using R pheatmap (<https://rdrr.io/cran/pheatmap/>) or EnhancedVolcano package (<https://github.com/kevinblighe/EnhancedVolcano>) with *avg log2FC* > 0.25 and *pct* > 0.5 thresholds. Pseudobulk analysis using PseudobulkExpression function was performed on either normalized or normalized and scaled data. DotPlot and VlnPlot were used to assessed gene expression level (Average and % of expression) per cluster.

Sample preparation, immunostaining, and gating strategy of *Tgfr1*l/l *cdh5*CreERT2. Whole brains of P12 *Tgfr1*l*EKO* and littermate control (*Tgfr1* l/l) (*n* = 2 brains per replicate, 2 replicates per genotype, 4 datasets in total) were enzymatically digested using Neural Tissue Dissociation Kit (P) (Miltenyi Biotec, 130-092-628) for 30 min at 37 °C. During the digestion process, tissues were homogenized two times by up and down pipetting. The samples were put in the ice with 5% SVF DMEM to stop the enzymatic reaction and pass through a 40 µm cell strainer to remove cell debris (Falcon, 352340). After a PBS step washing, the myelin and the cell debris were separated from the pelleted cells by using the debris removal solution (Miltenyi Biotec, 130-109-398) and the cells were washed with PBS before proceeding to the red blood cell lysis (Miltenyi Biotec, 130-094-183). After 30 min blocking of 5 µg/mL of CD16/CD32 Fc Block (eBiosciences, 16-0161-82), the cells were stained at RT for 40 min with a combination of the following antibodies (PE anti-mouse CD140b Antibody (Biolegend, 136006); APC anti-mouse CD31 (BD Biosciences, 551262); FITC anti-mouse CD45 (Biolegend, 103108)). After washing the cells with PBS, the cells were stained 20 min at room temperature with live/dead fixable Aqua dead (ThermoFisher Scientific, L34957), allowing the staining of dead cells. After a washing step, the stained cells were resuspended with 2 mM of EDTA and 0.04 % of BSA. The setting up of the gates and compensations were allowed with the unstained cells, the single positive staining, the Fluorescence Minus One (FMO) and the isotype control staining (PE Rat IgG2a, κ Isotype control antibody

(BD Biosciences, 557229); APC Rat IgG2a, κ (BD Biosciences, 551139) and FITC Rat IgG2b, κ (Biolegend, 400633)). After removing the dead cells and doublets, 10 000 endothelial cells (CD45⁻ CD31⁺ CD140b⁻) and perivascular cells (CD45⁻ CD31⁻ CD140b⁺) were sorted with a BD FACS Aria Fusion and pooled for each sample.

Library preparation of *Tgfb β 1*/l *cdh5CreERT2* single cell suspension. Single-cell suspensions of brain cells were resuspended in PBS containing 0.04% ultra-pure BSA. scRNAseq libraries were prepared using the Chromium Single Cell Reagent Kits v3.1 (10x Genomics; Pleasanton, CA, USA) according to the manufacturer's instructions. The target recovery for each library was 10,000 cells. Generated GEX libraries were sequenced on an Illumina NovaSeq6000 S4, followed by de-multiplexing and mapping to the mouse genome (mm10-3.0.0) using Cell Ranger (pipeline version 3.1.0) (10x Genomics).

Data processing and EC clustering of the scRNAseq of *Tgfb β 1*/l *cdh5CreERT2*. Further processing and analysis of the 4 scRNAseq dataset were performed using R (version 4.4.0) as previously described²². For each individual dataset, the filtered gene expression matrix was analyzed using tools from the Seurat R-package v5.1.0⁶². Additional quality control steps filtered out genes that were expressed in less than 3 cells and cells that expressed fewer than 100 genes. Lastly, only cells which had fewer than 30000 transcripts, fewer than 10000 genes, and less than 15% of transcripts originating from mitochondrial genes were retained for further analysis. All 4 datasets were merged and normalized gene expression data, generated using the SCTransform function (regressing on nFeature_RNA and percent.mt), was used for dimensionality reduction and clustering, while differential gene expression analysis was performed on log-normalized RNA assay. We could not observe any decrease in the mRNA level of *Tgfb β 1* in the ECs isolated from *Tgfb β 1**EKO* retinas. This is because the 10X genomics kit captures the polyA mRNA and generates libraries from the 3' UTR, suggesting that the exon deletion does not decrease the total mRNA level, but nevertheless generates a loss-of-function *Tgfb β 1* mutation²². After subclustering of EC clusters (based on EC markers) from P12 Control and P12 *Tgfb β 1**EKO* datasets, principal components analysis was conducted on the SCT normalized expression matrix using the RunPCA function. Dimensional reduction of the data was performed using Uniform Manifold Approximation and Projection (UMAP)⁶⁴ as implemented in the *RunUMAP* function, where the number of PCA dimensions used for the *dims* parameter was 15, and all other settings were default. Cells were clustered from the resulting UMAP 2 dimensions using the FindClusters function with a resolution of 1 with all other parameters set to their default values. Cell clusters were annotated based on well-known marker gene expression, namely, EC clusters were identified by the presence of *Cldn5* and *Pecam1* expression while EC subtype were defined using markers as previously published^{13,21,22}. Pathway activity score was calculated with the VISION package⁶⁵ on SCT data using the msigdb.v2023.1.Mm pathway database and a collection of expert-annotated gene sets generated in-house⁶⁶, then added to the Seurat object as a new assay (VISION). Differential gene expression analysis between each annotated EC subtype was conducted using the FindMarkers function with default parameters, and DEGs were visualized using EnhancedVolcano package (<https://github.com/kevinblighe/EnhancedVolcano>) with avg log2FC > 0.25 and pct > 0.5 thresholds. Pseudobulk analysis using PseudobulkExpression function was performed on either normalized or normalized and scaled data. DotPlot and VlnPlot were used to assessed gene expression level (Average and % of expression) per cluster.

Nichenet analysis–neurovascular-unit communications. The differential Brain-Vascular interactome at each time point (P6, P12, Adult) was determined using NicheNet analysis (version 2.2.1)²⁴ comparing

thalamic and cortical regions. Transcripts (SCT normalized) expressed in at least 10% of ECs (scRNAseq, whole ECs used as receiver) or 25% of spots (Visium, CTX and THAL regions used as sender) were used to screen for NVU communications between brain regions and endothelial cells. Ligand-receptor pairs were then identified using a ligand-receptor and ligand-target database with a weighted network created by Browaeys et al.²⁴. Brain region ligand enrichment was determined using differential gene expression analysis (FindMarker) to determine regionally enriched genes as previously described²⁴. Predicted ligand activity was calculated using up-regulated genes from receivers ($p_{val_adj} \leq 0.05$ and $avg_log2FC \geq 0.25$) and ligands from senders ($p_{val_adj} \leq 0.05$ & $avg_log2FC \geq 1$) and the top 30 ligands (based on corrected area under the precision-recall curve, plotted on heatmap) with their respective receptors were visualized on Circos plot and the weight of every ligand-receptor interactions based on the ligand-receptor network were saved in Supplementary Data 3. Using the ligand and downstream target matrix provided by Browaeys et al., region-specific ligand activity at each time point was predicted utilizing the Pearson correlation of ligand expression and intracellular downstream gene expression from ECs. A higher Pearson correlation indicated a higher relationship between ligand expression from brain regions and intracellular gene activity from ECs, which further indicated higher ligand activity. For each time point, common interactions have been determined using ligands and receptors expressed above the 10% threshold in all brain regions and ECs, subtracted from THAL and CTX enriched ligand-receptors as performed above.

Cell culture

Primary Human Brain microvascular Endothelial Cells (HBECs, Cell systems, ACBRI 376) were cultured between passages 1 to 6 in EBMTM Basal Medium (Lonza CC-3121) supplemented with SingleQuotsTM Supplements (Lonza CC-4133). Cells were transfected with siRNA Control (Dharmacon, D-001810-10-05) or siRNA *Tgfb β 1* (Dharmacon, L-003929-00-0005) using Lipofectamine RNAiMAX (Invitrogen, #13778150). 24 h after transfection, HBECs were treated with 5 ng/mL of human recombinant TGF β 2 (cell-signaling, #8406) for 48 h, followed by 3 h starvation in DMEM media (ThermoFisher Scientific, 10569044). Next, transfected cells were treated or not by 10 min or 30 min treatment with 50 ng/mL of human recombinant VEGF-C (R&Dsystems, #9199-VC).

mBEC isolation and culture

Whole mouse brains were minced and digested for 1 h at 37 °C in collagenase/dispase (Sigma, 11097113001) supplemented with DNase I (50 μ g/mL, DN25) and TCLK (0.147 μ g/mL, T7254). Myelin and debris were removed as previously described, and endothelial cells were isolated by CD31-based magnetic purification. CD31 antibody (BD Pharmingen, 550274) was incubated overnight at 4 °C with Dynabeads (ThermoFisher Scientific, 11035), washed using a DynaMagTM-15 separator (ThermoFisher Scientific, 12301D), and resuspended in PBS–0.01% BSA. Cell suspensions were incubated with 100 μ L of CD31-coated beads for 30 min at room temperature under agitation, followed by magnetic washing and collection of CD31⁺ cells. Purified mBECs were seeded on collagen I-coated plates (ThermoFisher Scientific, A1064401) and cultured in EBM-2 basal medium (Lonza, CC-3121) supplemented with EGM-2 SingleQuots (Lonza, CC-4133) and 20% FBS (ThermoFisher Scientific, 12484028).

Western blot

Cells were lysed in Laemmli 1x (Sigma Aldrich, S3401). Protein lysates were loaded in 4–15% Criterion precast gel (Bio-Rad, #5671084), next transferred on nitrocellulose membrane (Bio-Rad, 1620112), blocked for 1 h with 5% BSA in TBST, and then incubated ON at 4 °C in primary antibodies. After washing in TBST, membranes were incubated for 1 h

at RT with horseradish peroxidase-conjugated secondary antibodies. Antibody staining was revealed with SuperSignal™ West Femto Maximum Sensitivity Substrate (Thermo Fisher Scientific, 34096) or clarity western ECL substrate (Bio-Rad, 1705060S) using ChemiDoc MP imaging system (Bio-Rad). Band intensity was quantified using Image Lab software (Bio-Rad).

Metabolomic

CTX and THAL areas were micro-dissected from *Tgfb β 1^{l/l}* ($n = 9$) and *Tgfb β 1^{EKO}* ($n = 9$) brains and fast frozen until processed for amino acid quantification. Tissues were reduced to powder before extraction (from 50 mg) with 70% methanol and 1 mol/L hydroxylamine (at pH 7.6). Isotope-labeled internal and external standards were added as follows: 150 nmol $^{13}\text{C}_3$ -alanine, 50 nmol $^{13}\text{C}_2$ -glycine, 20 nmol $^{13}\text{C}_5$ -valine, 10 nmol $^{13}\text{C}_6$, ^{15}N -leucine, 10 nmol $^{13}\text{C}_6$ -isoleucine, 5 nmol $^{13}\text{C}_5$, ^{15}N -proline, 15 nmol $^{13}\text{C}_5$ -methionine, 50 nmol $^{13}\text{C}_5$, ^{15}N -serine, 50 nmol $^{13}\text{C}_4$, ^{15}N -threonine, 15 nmol D₅-phenylalanine, 100 nmol $^{13}\text{C}_4$, ^{15}N -aspartic acid, 400 nmol $^{13}\text{C}_5$, ^{15}N -glutamic acid, 30 nmol $^{13}\text{C}_6$ -arginine, 10 nmol $^{13}\text{C}_9$ -tyrosine, 20 nmol $^{13}\text{C}_6$ -histidine, 400 nmol $^{13}\text{C}_5$, $^{15}\text{N}_2$ -glutamine, 15 nmol $^{13}\text{C}_4$, $^{15}\text{N}_2$ -asparagine, 5 nmol $^{13}\text{C}_{11}$, $^{15}\text{N}_2$ tryptophan, 40 nmol $^{13}\text{C}_3$, ^{15}N cysteine, 50 nmol $^{13}\text{C}_6$, $^{15}\text{N}_2$ lysine. After 2 minutes in a sonication bath, samples were homogenized with 2.8 mm zirconium oxide beads (Omni International, Kennesaw, GA) in a Bead Ruptor homogenizer. Then, 1 mol/L hydrochloric acid was added (to reach a pH between 5 and 6), and samples were incubated at 70 °C for 15 min and then centrifuged at 22,000 $\times g$ for 10 min. The supernatants were collected and evaporated to near dryness. The samples were then rinsed once with 100% methanol and dried twice with ammonium sulfate. After a quick centrifugation at 7000 $\times g$, the supernatant was evaporated again to near dryness before transfer to GC-MS vials before evaporation to dryness. Finally, samples were solubilized in 100% pyridine at 45 °C for 90 min and followed by derivatization using *N*-methyl-*N*-tertbutyldimethylsilyltrifluoroacetamide at 90 °C for 4 h. Samples were injected into an Agilent 6890 N chromatograph coupled to a 5975 N mass spectrometer operated in electronic ionization (EI) mode (with helium as carrier gas) at a flow rate maintained throughout (7.7 mL/min) in split mode (5.6:1). The temperature program was fixed as follows: 150 °C for 3 min, increment of 7 °C/min up to 210 °C and maintained constant for 3 min, increment of 7 °C/min up to 310 °C and maintained for 5.5 min and then 40 °C/min up to 320 °C. Metabolites were identified according to their *m/z* and retention time and quantified using internal or external standards as well as standard curves.

Statistical analysis

Investigators were blinded during most data collection. Image quantifications and transcriptomic analyses were performed using automated pipelines to minimize bias. Sample sizes were determined based on previous studies of postnatal brain angiogenesis and on pilot experiments demonstrating reproducible effect sizes.

Animals from different litters were randomly assigned to experimental groups when possible. All results are represented as mean $\pm$ standard error of the mean (s.e.m). All statistical analysis and graph images were performed using GraphPad Prism 9 software. No statistical method was used to pre-determine the sample size. For multiple comparisons, ordinary ANOVA tests, Brown-Forsythe and Welch ANOVA tests, or Kruskal-Wallis nonparametric tests were used depending on the Gaussian distribution and equal or different variances between groups. For two groups, unpaired t-test, unpaired t-test with Welch's correction, or Mann-Whitney test were used depending on equal or different variances between groups. $P < 0.05$ was considered statistically significant. In Fig. 1e, g, P40 data equal to 0 were excluded to perform the statistical test. Single-cell and spatial transcriptomic differential expression analysis were carried out using a

Wilcoxon Rank-Sum test (the default test for the FindMarkers and FindAllMarkers functions in Seurat), which adjusts the *p*-values using the Bonferroni correction.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data that support the findings of this study are available within the article, its Supplementary Information, or in the public repositories listed below. The spatial transcriptomics and single-cell RNA sequencing (scRNAseq) datasets generated in this study have been deposited in NCBI's Gene Expression Omnibus (GEO) under accession codes: [GSE242215](#), [GSE242214](#), [GSE242213](#). The metabolomics dataset generated in this study has been deposited in the Metabolomics Workbench under accession code [ST004781](#). All source data underlying graphs are provided as a Source Data file. Source data are provided with this paper.

Code availability

Custom scripts used for transcriptomic analysis are available from the corresponding author upon request.

Materials availability

Materials are available from the corresponding author upon request.

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Author contributions

M.B. and A.D. conceived the project and wrote the manuscript. M.B. and A.D. designed and analyzed all experiments. A.D. supervised the project. M.B., E.D., and B.B. performed experiments. J.H., M.X.C., F.I., and G.C. performed computational analysis. S.L. performed 10x Genomics single-cell sequencing. I.R. and M.R. performed the metabolic experiment and analysis. F.L., J.S.J., and G.A. provided equipment, advice, and discussed results. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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